

# A brief introduction to mixing time in Markov chains

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# Plan of the talk

## 1. Markov chains

- Shuffling a deck of cards: how random is random?
- Definitions, equilibrium, convergence.

## 2. Mixing time and the cutoff phenomenon

- Distance to equilibrium, mixing time, spectral gap.
- Cutoff: the chain becomes random *abruptly*.

## 3. The simple exclusion process in dimension one

- $k$  indistinguishable particles on a segment.
- Sharp answer: cutoff at  $\frac{N^2 \log k}{2\pi^2}$ , and why.

Goal: everything is elementary; the questions are not.

# Part 1: Markov chains

# A question from everyday life

## Question

You shuffle a deck of 52 cards. How many shuffles until the deck is “random”?

To even ask the question we need:

- a **state space**: the  $52! \approx 10^{68}$  orderings of the deck;
- a **random rule** to pass from one ordering to the next (one shuffle);
- a notion of “**random**”: the uniform measure on the  $52!$  orderings;
- a way to **measure the distance** between the law of the deck after  $t$  shuffles and the uniform measure.

The first three items make a Markov chain; the fourth one leads to the mixing time.

# Markov chains: discrete time

$\Omega$ : a finite set (the **state space**).

## Definition

$P = (P(x, y))_{x, y \in \Omega}$  is a **transition matrix** if

$$P(x, y) \geq 0 \quad \text{and} \quad \sum_{y \in \Omega} P(x, y) = 1 \quad \forall x \in \Omega.$$

A random sequence  $(X_t)_{t \in \mathbb{N}}$  is a Markov chain with transition matrix  $P$  if

$$\mathbb{P}[X_{t+1} = y \mid X_t = x, X_{t-1}, \dots, X_0] = \mathbb{P}[X_{t+1} = y \mid X_t = x] = P(x, y).$$

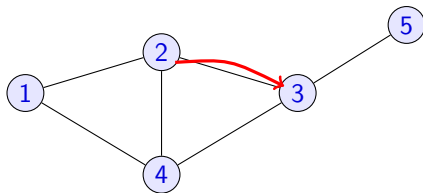
**The Markov property:** to predict the future, the present is enough — the past may be forgotten.

Iterating:  $\mathbb{P}[X_t = y \mid X_0 = x] = P^t(x, y)$ , the  $t$ -th power of the matrix.

## Example 1: random walk on a graph

$G = (V, E)$  a finite connected graph. From  $x$ , jump to a uniformly chosen neighbour:

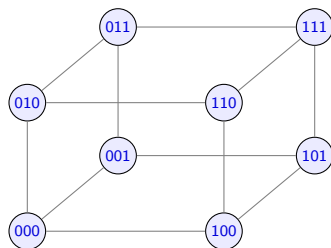
$$P(x, y) = \frac{1}{\deg(x)} \mathbf{1}_{\{x \sim y\}}.$$



- Card shuffling is a random walk on a graph whose vertices are the [52!](#) orderings.
- The [simple exclusion process](#) of Part 3 is a random walk on a graph with  $\binom{N}{k}$  vertices.

## Example 2: the hypercube (or: $n$ independent coins)

$\Omega = \{0,1\}^n$ ,  $|\Omega| = 2^n$ . At each step, pick a coordinate  $i \in \llbracket 1, n \rrbracket$  uniformly and **resample**  $x_i$  by a fair coin.



**Why this example?** It is a toy model where **everything** can be computed by hand, and it already displays the phenomenon (cutoff) we are after.

Equilibrium: the uniform measure on  $\{0,1\}^n$ .

## Continuous time: the generator

- **Generator:**  $\mathcal{L} = (r(x, y))_{x, y \in \Omega}$ , an  $\Omega \times \Omega$  matrix with
  - ▶ off-diagonal entries  $r(x, y) \geq 0$  (jump rates),
  - ▶ every row sum equal to zero.

For  $f : \Omega \rightarrow \mathbb{R}$ ,

$$(\mathcal{L}f)(x) := \sum_{y \in \Omega} r(x, y) (f(y) - f(x)).$$

- **Markov semi-group**  $(P_t)_{t \geq 0}$ :

$$P_t := e^{t\mathcal{L}} = \sum_{m \geq 0} \frac{(t\mathcal{L})^m}{m!}, \quad P_t(x, y) \geq 0, \quad \sum_{y \in \Omega} P_t(x, y) = 1.$$

**Picture:** on each oriented pair  $(x, y)$  put an independent Poisson clock of rate  $r(x, y)$ ; when it rings and the chain sits at  $x$ , it jumps to  $y$ .

Continuous time removes periodicity issues and makes formulas cleaner — we use it from now on.

# Equilibrium

- $\mu$  is an invariant probability measure if

$$(\forall t \geq 0, \mu P_t = \mu) \Leftrightarrow \mu \mathcal{L} = 0.$$

- Irreducible: for all  $x \neq y$  there is a path  $x = z_0, z_1, \dots, z_\ell = y$  with  $r(z_{m-1}, z_m) > 0$  for all  $m$ .

## Theorem (Convergence to equilibrium)

If  $(\Omega, \mathcal{L})$  is irreducible, there is a unique invariant probability measure  $\mu$ , and for every initial distribution  $\nu$ ,

$$\lim_{t \rightarrow \infty} \sum_{y \in \Omega} \left| \mathbb{P}^\nu [X_t = y] - \mu(y) \right| = 0.$$

This is qualitative. The whole talk is about making it quantitative:  
how long do we have to wait?

# Reversibility

- Detailed balance: for all  $x, y \in \Omega$ ,

$$\mu(x)r(x, y) = \mu(y)r(y, x) \quad \implies \quad \mu\mathcal{L} = 0.$$

(Much easier to check than  $\mu\mathcal{L} = 0$ : it is a local identity.)

- Then  $\mathcal{L}$  is self-adjoint in  $\langle f, g \rangle_\mu := \sum_x \mu(x)f(x)g(x)$ , and there is an orthonormal basis of eigenfunctions  $(\varphi_i)$ ,

$$\mathcal{L}\varphi_i = -\lambda_i\varphi_i, \quad 0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_{|\Omega|}, \quad P_t\varphi_i = e^{-\lambda_i t}\varphi_i.$$

## Moral

For a reversible chain, convergence to equilibrium is governed by the spectrum of  $\mathcal{L}$  — linear algebra enters the picture.

Example: random walk on a graph is reversible with respect to  $\mu(x) \propto \deg(x)$ .

# Why do we care?

- **Markov Chain Monte Carlo.** To sample from a complicated measure  $\mu$  (statistical physics, Bayesian statistics, combinatorial counting), build a chain with invariant measure  $\mu$  and run it.  
How long must we run it?  $\Rightarrow$  mixing time.
- **Statistical mechanics.** The relaxation of a physical system to equilibrium; phase transitions are often visible in the mixing time.
- **Card shuffling / combinatorics.** Random permutations, random walks on groups.
- **Algorithms.** Approximate counting (permanents, colourings, volumes) reduces to sampling, hence to mixing.

The mixing time is the running time of the algorithm.

## Part 2: Mixing time and cutoff

# Distance to equilibrium

- **Total variation distance** between two probability measures  $\alpha, \beta$  on  $\Omega$ :

$$\|\alpha - \beta\|_{\text{TV}} := \sup_{A \subset \Omega} |\alpha(A) - \beta(A)| = \frac{1}{2} \sum_{x \in \Omega} |\alpha(x) - \beta(x)|.$$

( $A$  = a statistical test;  $\|\alpha - \beta\|_{\text{TV}}$  = how well the best test distinguishes  $\alpha$  from  $\beta$ .)

- **Distance to equilibrium** (worst starting point):

$$d(t) := \max_{x \in \Omega} \|P_t(x, \cdot) - \mu\|_{\text{TV}}.$$

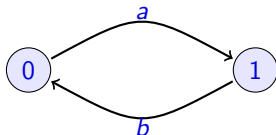
- $\varepsilon$ -mixing time: for  $\varepsilon \in (0, 1)$ ,

$$t_{\text{mix}}(\varepsilon) := \inf \{t \geq 0 : d(t) \leq \varepsilon\}, \quad t_{\text{mix}} := t_{\text{mix}}(1/4).$$

$d$  is non-increasing, and sub-multiplicative:  $d(s + t) \leq 2d(s)d(t)$ .

A chain we can solve by hand: two states

$\Omega = \{0, 1\}$ , rates  $r(0, 1) = a > 0$ ,  $r(1, 0) = b > 0$ .



- Equilibrium:  $\mu = \left( \frac{b}{a+b}, \frac{a}{a+b} \right)$ .
- Solving the  $2 \times 2$  linear system:  $P_t(0, 1) = \frac{a}{a+b} (1 - e^{-(a+b)t})$ , hence

$$d(t) = \frac{\max(a, b)}{a + b} e^{-(a+b)t}, \quad t_{\text{mix}}(\varepsilon) = \frac{1}{a + b} \log \frac{\max(a, b)}{(a + b)\varepsilon}.$$

- Eigenvalues of  $\mathcal{L}$ : 0 and  $-(a + b)$ .

Exponential decay at the rate given by the smallest nonzero eigenvalue.  
Keep this in mind.

# Spectral gap and relaxation time

Assume irreducibility and reversibility.

- **Spectral gap** = smallest nonzero eigenvalue of  $-\mathcal{L}$ :

$$\text{gap} := \inf_{f: \text{Var}_\mu(f) > 0} \frac{-\langle f, \mathcal{L}f \rangle_\mu}{\text{Var}_\mu(f)}, \quad \text{Var}_\mu(f) := \langle f, f \rangle_\mu - \langle f, \mathbf{1} \rangle_\mu^2.$$

- **Relaxation time**:  $t_{\text{rel}} := 1/\text{gap}$ .

## Comparison

With  $\mu_{\min} := \min_{x \in \Omega} \mu(x)$ , for  $\varepsilon \in (0, 1)$ :

$$t_{\text{rel}} \log \frac{1}{2\varepsilon} \leq t_{\text{mix}}(\varepsilon) \leq t_{\text{rel}} \log \frac{1}{2\varepsilon \mu_{\min}}, \quad \lim_{t \rightarrow \infty} \frac{1}{t} \log d(t) = -\text{gap}.$$

The gap between the two bounds is a factor  $\log(1/\mu_{\min}) \approx \log |\Omega|$  — **usually huge**. Which one is the truth?

# Sequences of chains

A single finite chain always mixes; the question is only interesting asymptotically.

Consider a sequence  $(\Omega_n, \mathcal{L}_n, \mu_n)_{n \in \mathbb{N}}$  with  $|\Omega_n| \rightarrow \infty$ , and write  $t_{\text{mix}}^{(n)}(\varepsilon)$ ,  $d_n(t)$ .

Q: how does  $t_{\text{mix}}^{(n)}(\varepsilon)$  grow with  $n$ , and how does it depend on  $\varepsilon$ ?

- Hypercube ( $n$  coordinates, each resampled at rate 1):  $t_{\text{mix}}^{(n)} \approx \frac{1}{2} \log n$ ,  $t_{\text{rel}} = 1$ .
- Random walk on the cycle  $\mathbb{Z}_n$ :  $t_{\text{mix}}^{(n)} \asymp n^2$ ,  $t_{\text{rel}} \asymp n^2$ .

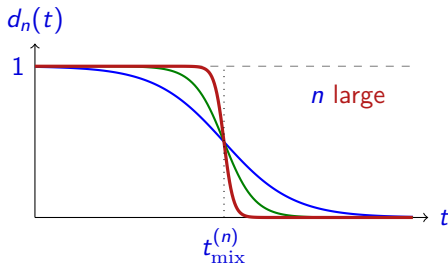
In the first case  $t_{\text{rel}} \ll t_{\text{mix}}$ ; in the second they are of the same order. This difference is exactly the cutoff phenomenon.

# The cutoff phenomenon

Definition (Aldous–Diaconis, 1986)

The sequence exhibits **cutoff** if for all  $\varepsilon \in (0, 1)$ ,

$$\lim_{n \rightarrow \infty} \frac{t_{\text{mix}}^{(n)}(\varepsilon)}{t_{\text{mix}}^{(n)}(1 - \varepsilon)} = 1 \quad \Longleftrightarrow \quad \lim_{n \rightarrow \infty} d_n \left( c t_{\text{mix}}^{(n)} \right) = \begin{cases} 1 & \text{if } c < 1, \\ 0 & \text{if } c > 1. \end{cases}$$



The profile becomes a **step function**: the chain is far from equilibrium, and then suddenly it is not.

## Cutoff on the hypercube: the proof in one slide

$\Omega_n = \{0, 1\}^n$ ; each coordinate has an independent Poisson clock of rate 1 and is resampled by a fair coin when it rings. Start from  $x = (0, \dots, 0)$ .

- **Lower bound.** Count  $W(x) = \sum_{i \leq n} x_i$ . A coordinate is still un-refreshed with probability  $e^{-t}$ , so

$$\mathbb{E}^x[W_t] = \frac{n}{2} - \frac{n e^{-t}}{2}, \quad \text{Var}(W_t) \leq \frac{n}{4}, \quad \mathbb{E}^\mu[W] = \frac{n}{2}, \quad \text{Var}_\mu(W) = \frac{n}{4}.$$

The bias  $\frac{n e^{-t}}{2}$  beats the noise  $\sqrt{n}$  as long as  $t \leq \frac{1}{2} \log n - c$ : Chebyshev gives  $d_n(t) \rightarrow 1$ .

- **Upper bound.** The chain is a product of  $n$  two-state chains, so a Fourier ( $L^2$ ) computation gives

$$4 d_n(t)^2 \leq (1 + e^{-2t})^n - 1 \xrightarrow[t = \frac{1}{2}(\log n + c)]{} e^{e^{-c}} - 1.$$

Cutoff at time  $\frac{1}{2} \log n$ , with a window of size  $O(1)$ .

## Back to the cards

### Riffle shuffle (Bayer–Diaconis, 1992)

For a deck of  $n$  cards the riffle shuffle has cutoff at  $\frac{3}{2} \log_2 n$  shuffles. For  $n = 52$ : about 7 shuffles.

### Random transpositions (Diaconis–Shahshahani, 1981)

Pick two cards uniformly and exchange them: cutoff at  $\frac{1}{2} n \log n$ .

### Top-to-random (Aldous–Diaconis, 1986)

Move the top card to a uniform position: cutoff at  $n \log n$ , by a coupon-collector argument.

**Common feature:** a sharp threshold, and a constant that can be identified. Cutoff was discovered precisely in this context.

# When does cutoff hold?

## Product condition

$$\text{gap}_n \cdot t_{\text{mix}}^{(n)} \longrightarrow \infty \quad \Longleftrightarrow \quad t_{\text{rel}}^{(n)} \ll t_{\text{mix}}^{(n)}.$$

- Cutoff  $\implies$  product condition (for reversible chains).
- The converse is **false** in general: there are counterexamples (see Levin–Peres, Ch. 18).
- It is **true** in several natural classes: birth-and-death chains (Ding–Lubetzky–Peres), trees, random walks on some families of graphs ...
- **Diaconis' question (1996): find a general criterion.** Still open.

Consequence: proving cutoff for a given model is usually a **model-by-model** affair, and requires sharp bounds on both sides.

# The standard toolbox

Upper bounds (the chain is close to  $\mu$ ):

- **Coupling.** Run two copies  $(X_t), (Y_t)$  with  $Y_0 \sim \mu$ , each marginally correct. Then

$$d(t) \leq \mathbb{P}[X_t \neq Y_t].$$

- Strong stationary times;  $L^2$  / spectral bounds; censoring (Peres–Winkler); path/log-Sobolev arguments.

Lower bounds (the chain is still far from  $\mu$ ):

- Find a **distinguishing statistic**  $f : \Omega \rightarrow \mathbb{R}$  with

$$\mathbb{E}^x[f(X_t)] \text{ far from } \mathbb{E}^\mu[f], \quad \text{fluctuations of } f \text{ under both small.}$$

Chebyshev then separates the two laws.

- **Wilson's method** (2004): take  $f$  = eigenfunction of  $\mathcal{L}$  with eigenvalue  $-\lambda$ ; then  $\mathbb{E}^x[f(X_t)] = e^{-\lambda t} f(x)$  decays at a known rate.

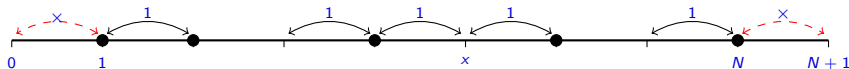
Part 3 puts these two ideas to work.

# Part 3: The simple exclusion process

in dimension one

# The simple exclusion process (SEP)

$k$  indistinguishable particles on the segment  $\llbracket 1, N \rrbracket$ .



- (A) Each site holds at most one particle (the exclusion rule); the number of particles is conserved.
  - (B) On each edge  $\{x, x+1\}$ ,  $1 \leq x < N$ , place an independent Poisson clock of rate 1. When it rings, swap the contents of the two sites.
- (No clock on  $\{0, 1\}$  and  $\{N, N+1\}$ : the particles stay in the segment.)

# Setup

- State space (1: particle, 0: empty site):

$$\Omega_{N,k} := \left\{ \xi : \llbracket 1, N \rrbracket \rightarrow \{0, 1\} \mid \sum_{x=1}^N \xi(x) = k \right\}, \quad |\Omega_{N,k}| = \binom{N}{k}.$$

- Generator: for  $f : \Omega_{N,k} \rightarrow \mathbb{R}$ ,

$$(\mathcal{L}_{N,k} f)(\xi) := \sum_{x=1}^{N-1} [f(\xi \circ \tau_{x,x+1}) - f(\xi)],$$

where  $\tau_{x,x+1}$  is the transposition of  $x$  and  $x+1$ .

- The uniform measure  $\mu_{N,k}$  on  $\Omega_{N,k}$  satisfies detailed balance,

$$\mu_{N,k}(\xi) \mathcal{L}_{N,k}(\xi, \eta) = \mu_{N,k}(\eta) \mathcal{L}_{N,k}(\eta, \xi),$$

hence it is the (unique) invariant probability measure.

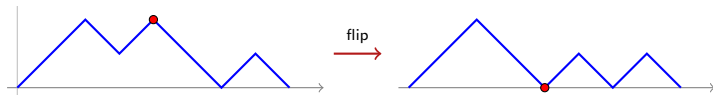
Question: starting from the worst configuration, how long until the particles are “uniformly placed”?

# Three faces of the same chain

- $k = 1$ : a single particle = random walk on  $\llbracket 1, N \rrbracket$ .
- **Labelled particles** ( $k = N$ , all distinct): the **interchange process**, i.e. the adjacent transposition shuffle — shuffling a deck by exchanging neighbouring cards.
- **Height function**  $h : \llbracket 0, N \rrbracket \rightarrow \mathbb{Z}$ ,

$$h(0) = 0, \quad h(x) = \sum_{y=1}^x (2\xi(y) - 1).$$

SEP becomes the **corner-flip dynamics** of a lattice path.



A particle/hole swap = a local maximum turning into a local minimum.

## Step 1: the spectral gap

**One particle** ( $k = 1$ ): the generator is the discrete Laplacian (Neumann b.c.). Everything is explicit:

$$g_i(x) = \cos\left(\frac{i\pi(x - 1/2)}{N}\right), \quad \mathcal{L}_{N,1} g_i = -\lambda_i g_i, \quad \lambda_i = 2(1 - \cos(i\pi/N))$$
$$\text{gap}_{N,1} = 2(1 - \cos(\pi/N)) = (1 + o(1))\frac{\pi^2}{N^2}.$$

Aldous' spectral gap conjecture (Caputo–Liggett–Richthammer, JAMS 2010)

For the interchange process on any finite weighted graph, the spectral gap equals that of the **single-particle** random walk. On the segment:

$$\text{gap}_{N,k} = \text{gap}_{N,1} = (1 + o(1))\frac{\pi^2}{N^2}, \quad \forall k \in \llbracket 1, N-1 \rrbracket.$$

So  $t_{\text{rel}} \approx N^2/\pi^2$ , independently of the number of particles.

## Step 2: the mixing time

The general comparison only gives

$$\frac{N^2}{\pi^2} \log \frac{1}{2\varepsilon} \lesssim t_{\text{mix}}^{N,k}(\varepsilon) \lesssim \frac{N^2}{\pi^2} \cdot N.$$

Theorem (Lacoin, Ann. Probab. 2016)

If  $\min(k, N - k) \rightarrow \infty$ , then for every  $\varepsilon \in (0, 1)$ ,

$$t_{\text{mix}}^{N,k}(\varepsilon) = (1 + o(1)) \frac{N^2 \log k}{2\pi^2},$$

so the simple exclusion process on the segment exhibits **cutoff**.

- The truth is the lower bound **times  $\log k$** , not times  $N$ .
- Earlier:  $t_{\text{mix}} \asymp N^2 \log N$  (Wilson 2004); general graphs (Oliveira 2013).
- Same statement for the **adjacent transposition shuffle** (labelled particles).

Where does  $\frac{N^2 \log k}{2\pi^2}$  come from?

A two-line heuristic, via the height function  $h_t$ .

(i) **Averages follow the heat equation.** The generator acts on  $h$  as the discrete Laplacian, so

$$\partial_t \mathbb{E}[h_t] = \Delta \mathbb{E}[h_t] \implies \text{first Fourier mode decays like } e^{-\pi^2 t / N^2}.$$

(ii) **Worst start.** All  $k$  particles packed on the left: the initial height profile has amplitude of order  $k$ .

(iii) **Equilibrium fluctuations.** Under  $\mu_{N,k}$  the height function fluctuates on the scale  $\sqrt{k}$ .

Mixing occurs when the deterministic profile shrinks below the equilibrium noise:

$$k e^{-\pi^2 t / N^2} \approx \sqrt{k} \iff t \approx \frac{N^2 \log k}{2\pi^2}.$$

The factor  $\frac{1}{2}$  is exactly “signal  $k$  versus noise  $\sqrt{k}$ ”.

## Making the lower bound rigorous

Test function built from the first eigenfunction of the one-particle walk:

$$\Phi(\xi) := \sum_{x=1}^N g_1(x) \xi(x), \quad g_1(x) = \cos\left(\frac{\pi(x - 1/2)}{N}\right).$$

- Exact decay: since  $\mathcal{L}_{N,1} g_1 = -\text{gap } g_1$ , the particle density evolves linearly and

$$\mathbb{E}^\xi [\Phi(\xi_t)] = e^{-\text{gap } t} \Phi(\xi).$$

Starting with the particles packed on the left,  $\Phi(\xi_0) \asymp k$ .

- Second moment:  $\text{Var}(\Phi) = O(k)$ , both along the dynamics and under  $\mu_{N,k}$  (negative correlations help).
- Chebyshev: as long as  $e^{-\text{gap } t} k \gg \sqrt{k}$ , the events  $\{\Phi > a\}$  separate  $P_t^\xi$  from  $\mu_{N,k}$ , so  $d_{N,k}(t) \approx 1$ .

$$\implies t_{\text{mix}}^{N,k}(\varepsilon) \geq (1 - o(1)) \frac{N^2 \log k}{2\pi^2}. \quad (\text{This is Wilson's method.})$$

# Making the upper bound rigorous

Harder: one must show the chain **is** mixed, not just that a statistic is.

- **Monotone coupling**. The height functions are partially ordered and the dynamics preserves the order. It is enough to couple the two **extremal** initial conditions  $h^{\max}$  and  $h^{\min}$ :

$$d_{N,k}(t) \leq \mathbb{P} [h_t^{\max} \neq h_t^{\min}] .$$

- **Area argument** (Wilson). The area  $A_t = \sum_x (h_t^{\max}(x) - h_t^{\min}(x)) \geq 0$  is a supermartingale decaying at rate **gap**, so  $\mathbb{E}[A_t] \leq e^{-\text{gap} t} A_0$  becomes  **$o(1)$**  slightly after  $\frac{N^2 \log k}{2\pi^2}$ .
- **Censoring** (Peres–Winkler). Removing updates can only make the chain less mixed — a licence to run the dynamics in a convenient order (e.g. on sub-segments first).

**Coalescence then happens within  $o(N^2 \log k)$  extra time.**

# What happens if we perturb the model?

- **Asymmetric jumps (ASEP)**: particles prefer one direction. Mixing time of order  $N$  instead of  $N^2 \log N$ , with cutoff (Labbé–Lacoin).
- **Inhomogeneous / random environment**: put a conductance  $c(x, x+1) > 0$  on each edge, i.i.d. say.

$$\text{gap}_{N,k} = \text{gap}_{N,1}, \quad t_{\text{mix}}^{N,k}(\varepsilon) = (1 + o(1)) \frac{N^2 \log k}{2\pi^2}$$

under a law of large numbers for the resistances  $1/c$  [Y. '24+]. The homogenized medium behaves like the homogeneous one — but heavy tails, traps and slow bonds change the answer.

- **Higher dimension**, general graphs, other conservative dynamics (zero-range process, ...): largely open.

A general principle: “one particle determines the whole system” — when it is true, and when it fails.

# Summary

- A Markov chain on a finite state space converges to equilibrium; **the mixing time** says how fast.
- Spectral gap gives the asymptotic rate, but can be off by a factor  $\log |\Omega|$ ; determining the constant requires real work.
- **Cutoff**: convergence happens abruptly. Proved case by case; a general criterion is an open problem.
- **Simple exclusion on  $[1, N]$** : gap  $= \pi^2/N^2$  (Aldous' conjecture), cutoff at  $\frac{N^2 \log k}{2\pi^2}$  — heat equation + equilibrium fluctuations.

## Suggested reading

- D. Levin, Y. Peres, *Markov Chains and Mixing Times*, 2nd ed., AMS 2017. (Chapters 1–6, 12, 18.)
- P. Diaconis, *The cutoff phenomenon in finite Markov chains*, PNAS 1996.
- T. Liggett, *Interacting Particle Systems*, Springer.

# Thank you!

Obrigado!    Xièxie!

Questions?